

PROJECT PROGRESS REPORT



Project31 - Longitudinal PET/CT Lesion Quantification and Treatment

Group Formal Tag: COMP3888_M17_03_ZL
Tutor Name: Zhenlin Liu
Client Name: Zilin Zhang

Group members details
(indicate if any member is doing multiple units)

No.	Student Name	Student ID	Student Unikey
1	Qiang Li	540788868	qili0179
2	Linkang Wen	540276619	lwen0131
3	Yuqi Wu	540035669	yuwu0501
4	Haoxiong Pan	530425128	hpan0285

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1 Introduction

This project aims to develop a reproducible research tool for reviewing and quantifying lesion changes in longitudinal PET/CT scans. In longitudinal cancer imaging, lesions within the same patient may respond differently over time: some may remain persistent, some may disappear, new lesions may appear, and multiple lesions may merge. Reviewing these changes manually can be difficult and time-consuming, particularly when scans contain multiple lesions and differ in patient positioning, image geometry, voxel spacing, and anatomical alignment.

The project therefore focuses on presenting lesion-level and patient-level changes in a clear and structured manner by comparing baseline and follow-up scans. The expected outcome is an end-to-end Python-based analysis pipeline capable of organising longitudinal scan pairs, extracting individual lesions and their quantitative features, aligning baseline and follow-up images, automatically matching lesions across timepoints, evaluating the matching results against expert correspondence data, and presenting the results through an interactive Streamlit interface. The system is designed for descriptive research review and does not provide clinical diagnosis, autonomous response classification, or treatment recommendations.

The key stakeholders are the project client, the COMP3888 development team, and prospective researchers or clinical-domain users who may review longitudinal imaging results. Regular client meetings and demonstrations are used to confirm project requirements, review technical decisions, and obtain feedback on registration, lesion matching, visualisation, and system usability. Client feedback has influenced the development of an interpretable rigid-registration baseline, automated lesion matching, and multiplanar visualisation.

The project uses two complementary public datasets. Cohort A, based on the autoPET-IV Longitudinal-CT dataset, provides expert-confirmed lesion correspondence between baseline and follow-up scans and is used primarily for developing and validating the lesion-tracking method. Cohort B is based on the PSMA subset of autoPET-III and is intended for broader PET/CT analysis. The main software resources include Python, NiBabel, SciPy, NumPy, Pandas, ITKElastix, NetworkX, Streamlit, GitHub, and Jira.

The main project risks include registration failure caused by differences in scan geometry or anatomy, unreliable lesion correspondence in complex cases such as lesion appearance, disappearance, splitting, or merging, incomplete PET information, and errors propagating from lesion extraction into later matching stages. The project mitigates these risks by first validating the pipeline on a smaller Cohort A subset, preserving intermediate results, testing modules independently, visually inspecting registration, and quantitatively comparing predicted lesion correspondences against expert ground truth. This follows the staged-risk approach defined in the

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original proposal, which recommended smaller representative cases before full-cohort processing and explicit handling of complex lesion-matching cases.

At the current progress-report stage, the team has completed a functional baseline pipeline covering data preparation, lesion extraction, rigid registration, aligned feature generation, min-cost-flow lesion matching, quantitative evaluation, and interactive visualisation. The next stage will focus on improving registration reliability, expanding evaluation coverage, and further integrating the analysis pipeline with the client-facing interface.

2 Major Achievements and Project Progress

The project has progressed from initial data-loading and visualisation experiments to an integrated baseline system for longitudinal lesion tracking. The current implementation covers patient-data preparation, medical-image loading, lesion component extraction, baseline-to-follow-up image registration, aligned lesion-feature generation, candidate-pair cost calculation, min-cost-flow lesion matching, quantitative evaluation against expert correspondence data, and interactive visualisation. These components have been developed as a connected pipeline rather than as isolated prototypes, allowing intermediate outputs to be inspected, tested, and reused across different stages of the system.

The development work has been guided by the requirements established in the original project proposal and refined through client meetings and demonstrations. In particular, the system is expected to organise longitudinal scans reproducibly, extract lesion-level information, align baseline and follow-up scans, identify lesion correspondence across timepoints, quantitatively evaluate the results, and provide an interface through which users can visually inspect the longitudinal changes. The following table summarises the relationship between the main user requirements, current achievements, and implementation evidence.

Client / User Requirement	Achievement and Current Progress	Evidence
Organise longitudinal patient data reproducibly	Implemented Cohort A patient discovery, scan manifest and BL/FU pair generation.	GitHub: tools/prepare_cohort_a_subset.py
Extract individual lesions from masks	Implemented 3D connected-component lesion extraction with centroid and volume calculation.	GitHub: src/lesion_components.py
Align baseline and follow-up scans	Implemented rigid BL→FU registration using ITKElastix and resampled BL masks.	GitHub: registration tool

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Automatically track lesions	Implemented candidate-cost generation and NetworkX min-cost-flow matching.	GitHub: matching module
Quantitatively evaluate tracking	Implemented comparison against expert Cohort A correspondence with accuracy, precision, recall and F1.	GitHub: evaluation tool
Visually inspect longitudinal results	Implemented Streamlit views for original BL, registered BL and FU images with lesion overlays and axial/coronal/sagittal viewing.	GitHub: Streamlit tool

A major early achievement was the development of a reproducible data-preparation workflow for Cohort A. The system discovers patient data and generates structured scan and baseline-follow-up pair manifests containing CT, PET, lesion-mask and expert-reference information. These manifests provide a consistent data-access layer for downstream processing and reduce dependence on manually configured file paths. This is particularly important for batch processing, where registration, feature extraction, matching and evaluation must operate on the same patient pairs.

The image-loading and lesion-extraction components were then developed to convert the imaging data into structured representations suitable for longitudinal comparison. Medical-image geometry and spatial metadata are preserved when loading NIfTI files. Binary lesion masks are separated into individual three-dimensional connected components, from which lesion identifiers, physical centroids and lesion volumes are calculated. These lesion-level records provide the basic representation required by later registration, feature-generation and matching stages. The current report already identifies the manifest, loading and lesion-extraction workflow as a completed part of the baseline system.

Another major achievement was the implementation of rigid longitudinal image registration. Direct comparison between baseline and follow-up lesions is unreliable when scans differ in patient positioning, voxel spacing, image geometry or anatomical alignment. The system therefore registers the baseline CT to the follow-up CT using ITKElastix and resamples the baseline lesion mask into the follow-up geometry.

The follow-up CT is treated as the fixed image so that subsequent lesion comparison can take place within a common follow-up coordinate system. Lesion positions are represented using physical/world coordinates rather than relying only on raw voxel indices, because voxel indices are not directly comparable when image geometry differs. Nearest-neighbour interpolation is used when resampling lesion masks to avoid introducing invalid fractional labels. The current implementation

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deliberately uses rigid registration as an interpretable baseline, consistent with the agreed project scope, rather than introducing complex deformable registration at this stage.

Registration outputs can also be reused during later processing rather than recalculating the same transformation for every experiment. This reduces repeated computation during aligned feature generation, lesion matching and evaluation, while also making the development workflow more reproducible.

Following registration, aligned lesion features are generated in a common coordinate system and used to calculate candidate baseline-to-follow-up relationships. The current candidate-cost formulation considers spatial distance and lesion-size difference, while the architecture allows additional PET-derived measurements to be incorporated when appropriate.

A key technical achievement was the formulation of lesion correspondence as a minimum-cost-flow optimisation problem. A simple nearest-neighbour strategy is insufficient for longitudinal lesion tracking because correspondence between two scans is not always strictly one-to-one. Lesions may remain persistent, disappear, newly appear, or participate in merge-related events. The min-cost-flow formulation therefore provides a more flexible optimisation framework for representing different longitudinal outcomes.

The team constructs the flow network, defines candidate edges, matching costs, capacities, unmatched-lesion penalties and merge-related slots. NetworkX is then used to solve the resulting minimum-cost-flow problem, after which the returned flow is decoded into explicit baseline-to-follow-up lesion correspondences. Separating network construction, optimisation and decoding also improves maintainability because cost formulations or event representations can be modified without replacing the optimisation solver itself.

This represents a progression beyond the simpler matching approach proposed at the beginning of the project. The original proposal required a transparent method capable of identifying persistent, new and disappeared lesions; the current implementation extends this concept into a global optimisation framework while retaining interpretable lesion-level costs and matching outcomes.

The project has also implemented a quantitative evaluation workflow using the expert-provided Cohort A lesion correspondences. Predicted lesion matches are compared with the expert reference data, and performance is reported using tracking accuracy, precision, recall and F1 score. Topology-classification performance is evaluated separately.

The current baseline has been evaluated on 11 Cohort A patients, achieving 91.0% tracking accuracy, 90.4% precision, 91.0% recall and 90.7% F1. Topology classification achieved 88.5% accuracy and 88.2% F1. Four additional patients from the original development subset are currently excluded from these results because of a known rigid-registration initialisation issue. Therefore, the reported results represent the performance of the currently evaluable baseline rather than final full-cohort performance.

Metric	Current Result
Tracking Accuracy	91.0%

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Precision	90.4%
Recall	91.0%
F1 Score	90.7%
Topology Accuracy	88.5%
Topology F1	88.2%

The evaluation stage has been useful not only for reporting quantitative performance but also for identifying weaknesses in upstream stages of the pipeline. In particular, the registration issue affecting four patients demonstrates that final matching performance depends on reliable spatial alignment before lesion correspondence is calculated. Evaluation has also revealed cases in which lesion extraction or expert-to-extracted-lesion mapping is incomplete or ambiguous. These cases are being treated as engineering issues for further investigation rather than being silently ignored during performance assessment.

In parallel with the processing pipeline, the team developed an interactive Streamlit prototype for visual inspection of longitudinal scans and registration results. The interface allows users to inspect the original baseline image, the rigidly registered baseline image and the original follow-up image. Lesion-mask overlays can be enabled or disabled to support visual validation of lesion positions and alignment.

The interface has also been extended beyond axial inspection to support axial, coronal and sagittal views. This enables longitudinal cases to be reviewed from multiple anatomical orientations and provides a more useful validation environment than inspecting numerical output alone. The visualisation therefore serves two purposes: it provides an early user-facing prototype and also acts as an engineering validation tool for identifying registration or lesion-mask problems that may not be apparent from tabular results. The current implementation already supports inspection of original images, registered images and lesion overlays.

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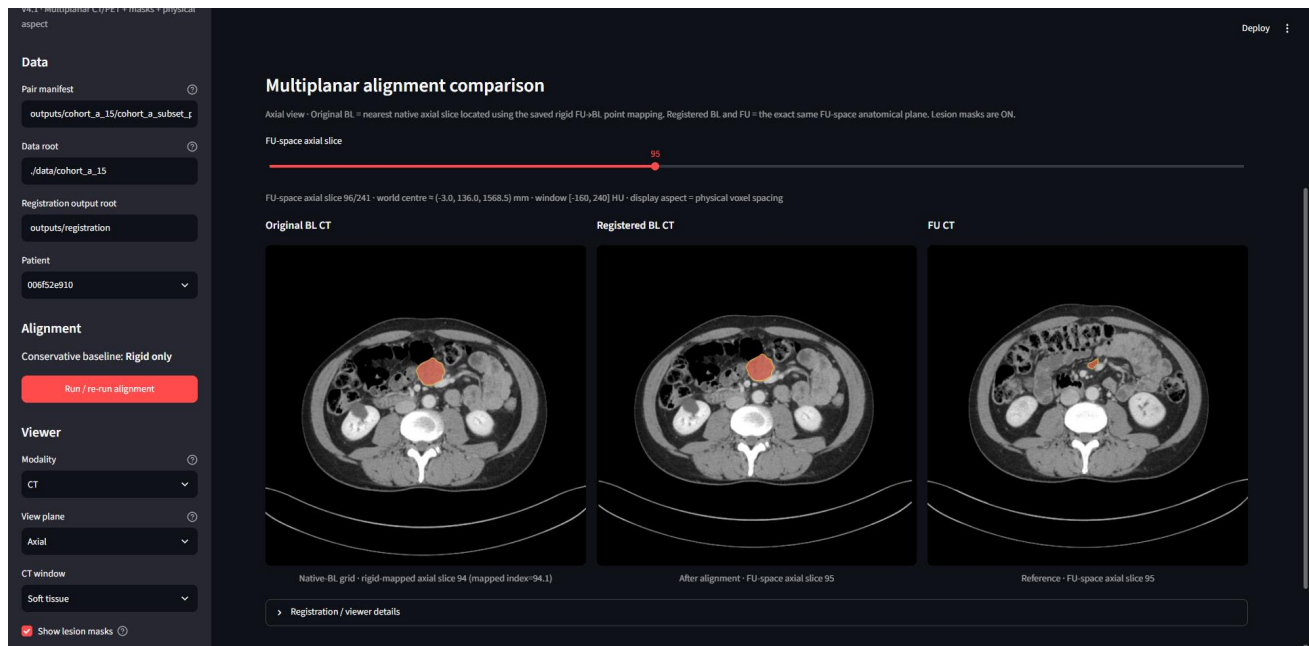


Figure 1. Interactive longitudinal imaging prototype showing the original baseline scan, rigidly registered baseline scan and follow-up scan with optional lesion-mask overlays.

Overall, the project has progressed beyond separate proof-of-concept scripts and now provides a functional end-to-end baseline for longitudinal lesion analysis. The current system can prepare patient data, extract individual lesions, align baseline and follow-up scans, generate comparable lesion features, perform optimisation-based lesion matching, evaluate predicted correspondences against expert reference data, and visually inspect the results through an interactive interface.

The remaining work is therefore focused primarily on improving the reliability and coverage of the existing system rather than replacing the core pipeline. In particular, the team will investigate the rigid-registration initialisation issue affecting the remaining Cohort A cases, extend quantitative evaluation after this issue is resolved, further validate matching parameters, and strengthen integration between the lesion-tracking pipeline and the final client-facing interface.

3 System Specification and Architecture Design

The system follows a modular longitudinal imaging architecture in which each processing stage has a clearly defined input and output. This design allows individual components to be tested, replaced, or improved without redesigning the complete system. It also supports reproducible debugging because intermediate outputs such as patient manifests, aligned lesion features, candidate-cost matrices, and predicted correspondences can be inspected independently.

The current end-to-end architecture is shown in Figure 2.

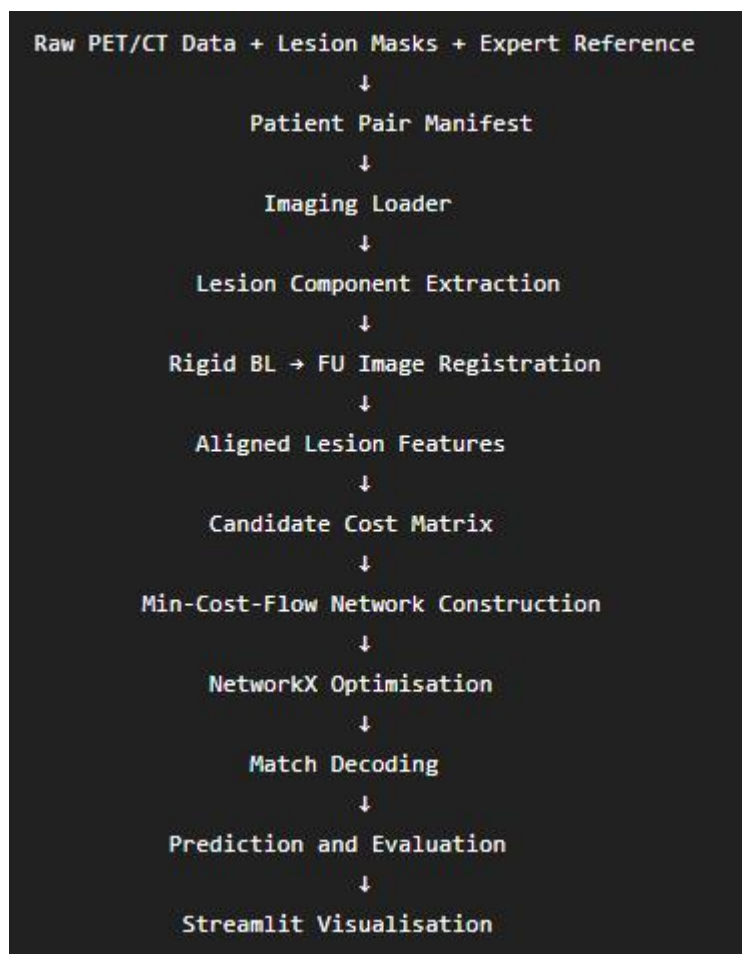


Figure 2. High-level architecture of the longitudinal lesion-tracking system.

This architecture extends the sequential workflow originally proposed for the project, which consisted of patient pairing, registration, feature extraction, lesion matching, change analysis, structured outputs, and Streamlit visualisation.

The patient manifest and imaging loader form the system's data-access layer. The manifest maintains a consistent association between baseline and follow-up CT, PET, lesion masks, and expert-reference

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files. The loader preserves image geometry, voxel spacing, affine information, and other spatial metadata required by later registration and lesion comparison. The lesion-component module then converts each binary lesion mask into separate three-dimensional lesion objects with stable identifiers, physical centroids, and lesion volumes.

A central architectural decision is to perform image registration before lesion matching. Baseline and follow-up scans may differ in image geometry, voxel spacing, orientation, and patient position; therefore, lesion distances calculated directly from native voxel coordinates would not provide a reliable basis for correspondence. The system registers the baseline CT to the follow-up CT and transforms the associated baseline lesion mask into the same follow-up geometry before candidate relationships are generated.

The follow-up CT is used as the **fixed image**, while the baseline CT is the moving image. This choice ensures that downstream lesion correspondence is represented in the same coordinate system as the follow-up examination. Lesion locations are compared using physical/world coordinates rather than raw voxel indices because world coordinates preserve spatial meaning across images with different resolutions or affine transformations. Nearest-neighbour interpolation is used when resampling lesion masks so that binary or discrete lesion labels are preserved instead of being converted into invalid fractional values. These decisions are already part of the implemented architecture.

The current system uses **rigid registration** rather than deformable registration. This was selected as an interpretable and conservative baseline and remains consistent with the agreed project scope, which explicitly excluded complex deformable registration. Rigid registration reduces the possibility of unrealistic local deformation and makes the transformation easier to inspect, although it cannot fully compensate for anatomical changes between longitudinal scans. Registration quality is therefore treated as an independent source of uncertainty and is visually inspected through the Streamlit interface.

After registration, the system generates an aligned lesion-feature representation. Each baseline and follow-up lesion can be described using spatial position, volume, and optional PET-related measurements. Candidate lesion pairs are then created and assigned costs according to differences between these features. The current baseline primarily uses spatial distance and lesion-size difference, while the architecture permits additional PET measurements to be incorporated without changing the overall matching pipeline.

Lesion correspondence is formulated as a **minimum-cost-flow optimisation problem** rather than a simple pairwise nearest-neighbour assignment. This design was chosen because longitudinal lesion behaviour may include persistent lesions, disappearing lesions, newly appearing lesions, and merge-related events. These relationships cannot always be represented reliably by independent one-to-one nearest-neighbour decisions.

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The matching architecture is deliberately separated into three stages:

1. **Network construction**, where candidate lesion relationships, costs, capacities, unmatched penalties, and merge-related slots are defined.
2. **Optimisation**, where NetworkX solves the resulting minimum-cost-flow problem.
3. **Decoding**, where the flow solution is translated into explicit baseline-to-follow-up lesion correspondences.

This separation makes the matching component easier to maintain and extend. For example, feature weights, unmatched penalties, candidate thresholds, or merge handling can be adjusted without changing the optimisation solver. Similarly, the solver can be replaced later without redesigning the upstream feature-generation stages.

The output of the matching stage is evaluated against the expert-provided Cohort A correspondence data. Importantly, expert ground truth is not used as an input to the matching algorithm; it is kept separate and used only after prediction for quantitative evaluation. This separation prevents information leakage and ensures that reported performance measures the behaviour of the implemented pipeline rather than reproducing known correspondence labels.

The final architecture also includes an interactive Streamlit visualisation layer. The interface is separated from the underlying processing modules so that registration and matching can be executed and tested independently of the user interface. The current viewer provides original baseline, registered baseline, and follow-up images, lesion-mask overlays, patient selection, image modality selection, CT windowing, and multiplanar inspection. This modular separation allows the same processing outputs to support both automated evaluation and interactive user review.

The main architectural decisions are summarised below.

Design Decision	Rationale
Modular pipeline	Allows components to be tested and replaced independently
Manifest-based data access	Ensures consistent patient and scan pairing across tools
Registration before matching	Places lesions into a comparable spatial coordinate system

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FU CT as fixed image	Keeps downstream outputs in follow-up geometry
Physical/world coordinates	Remain meaningful across scans with different voxel spacing and affine geometry
Nearest-neighbour mask interpolation	Preserves discrete lesion labels during resampling
Rigid registration	Provides an interpretable baseline consistent with project scope
Feature-based candidate costs	Makes matching decisions transparent and extendable
Min-cost-flow formulation	Supports global optimisation and non-trivial longitudinal lesion events
Separate construction / solving / decoding	Improves maintainability and future extensibility
Ground truth separated from prediction	Prevents information leakage during evaluation
Processing separated from Streamlit UI	Supports both automated evaluation and interactive review

Overall, the architecture is designed to prioritise **reproducibility, interpretability, modularity, and incremental extensibility**. The system can currently execute the complete longitudinal tracking workflow while still allowing individual components—particularly registration, candidate-cost design, merge handling, and visualisation—to be improved independently during the remaining project stages.

4 Discipline Knowledge and Tools

This project requires the integration of knowledge from medical imaging, image processing, spatial geometry, graph optimisation, data analysis, and software engineering. These areas are closely connected within the longitudinal lesion-tracking pipeline. For example, lesion correspondence depends on spatial distance, but such distances are only meaningful when image geometry is interpreted correctly and baseline and follow-up scans have first been aligned into a comparable coordinate system.

An important area of discipline knowledge is **medical-image representation and spatial geometry**. NIfTI images contain not only voxel intensity arrays but also information such as voxel spacing, orientation, and affine transformations. Therefore, the same voxel index in two longitudinal scans does not necessarily correspond to the same physical anatomical position. **NiBabel** is used to preserve this spatial information during image loading, while physical/world coordinates are used when calculating lesion positions. **NumPy** and **Pandas** support numerical processing and structured storage of manifests, lesion features, candidate relationships, and evaluation results.

The project also applies **three-dimensional image-processing knowledge** to convert binary lesion masks into individual lesion objects. A single lesion mask may contain multiple disconnected regions, so treating the complete mask as one lesion would make longitudinal correspondence unreliable. **SciPy connected-component labelling** is therefore used to separate masks into individual three-dimensional components, from which lesion identifiers, centroids, and volumes are calculated for downstream analysis.

Medical-image registration is required because baseline and follow-up scans may differ in patient position, scan geometry, voxel spacing, and anatomical alignment. The use of rigid registration follows the agreed project scope. After investigating suitable implementation tools, the team selected **ITKElastix**. The underlying elastix framework was developed for intensity-based medical-image registration and provides a modular design in which transformations, similarity measures, optimisation strategies, and interpolation methods can be configured independently. Its scriptable workflow also makes it suitable for reproducible processing of multiple longitudinal patient pairs (Klein et al., 2010). ITKElastix was therefore selected as an established implementation framework for the required registration process rather than as the source of the rigid-registration requirement itself.

The lesion-matching stage requires knowledge of **graph representation and optimisation**. Following the client's requirement to use a minimum-cost-flow approach, the team investigated suitable Python implementations and selected **NetworkX**. NetworkX provides flexible graph structures and established graph algorithms, while allowing nodes, edges, costs, and capacities to be defined for application-specific problems (Hagberg et al., 2008). This makes it suitable for representing lesion correspondence as a customised flow network.

Importantly, NetworkX acts as the **general-purpose optimisation solver rather than the complete lesion-tracking solution**. The team is responsible for candidate generation, network construction, edge-cost calculation, capacities, unmatched-lesion penalties, merge-related handling, and decoding the resulting flow into lesion correspondences. Using an established solver reduces unnecessary implementation risk while keeping the lesion-specific matching model transparent and modifiable.

The project also applies **software-engineering and experimental-evaluation knowledge**. **Pytest** is used for automated testing, while intermediate CSV outputs allow individual pipeline stages to be

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inspected independently. **GitHub** supports version control and collaborative integration, and **Jira** is used to manage user stories, development tasks, and identified defects. **Streamlit** was selected for the interactive prototype because it integrates directly with the Python processing pipeline and supports image viewing, lesion-mask overlays, patient selection, and multiplanar inspection without requiring a separate frontend architecture.

Overall, tool selection was based on a clear distinction between **project requirements and implementation decisions**. Rigid registration and minimum-cost-flow matching followed the agreed project requirements, while the team selected ITKElastix and NetworkX as suitable implementation tools after evaluating their capabilities. Established libraries are used for general-purpose operations, allowing the team to focus development effort on project-specific lesion representation, matching logic, evaluation, and interactive review.

5 Group Processes and Dynamics

The team followed an iterative and self-managed development process supported by weekly internal meetings, client meetings, Jira, GitHub, and shared documentation. Project Manager, Tracker, Doomsayer, and Client Liaison roles were rotated so that all members participated in planning, progress tracking, risk identification, and communication.

Jira was used to manage user stories, tasks, and bugs, while GitHub supported version control and integration of independently developed modules. Meeting minutes recorded key decisions, action items, responsible members, and deadlines. This provided traceable evidence of both technical progress and team coordination.

The modular development approach allowed members to work independently while using agreed intermediate formats for integration. This was important because registration, feature extraction, matching, and evaluation were strongly dependent on one another. When issues such as registration failures or lesion-extraction problems were identified, they were documented and incorporated into later development priorities rather than being handled informally.

Client feedback was used to refine requirements and prioritise a stable, interpretable baseline before adding more complex functionality. One key lesson was that upstream problems can significantly affect downstream work. In future sprints, the team will define clearer acceptance criteria and perform integration testing earlier to reduce cross-module delays.

Overall, the group maintained effective collaboration through clear task ownership, regular communication, version control, shared documentation, and structured issue tracking.

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◇ Individual Contributions, Responsibilities, and Work Split

Summary of Group Members' contributions

Group Member	Technical Contributions	Team Contributions	Status
Qiang Li	Lesion visualisation and aligned lesion-feature extraction	Meeting minutes, documentation and review	Completed
Linkang Wen	Cohort A subset preparation and patient manifest	Planning and integration support	Completed
Haoxiong Pan	Image loading, lesion extraction, pair costs and multiplanar views	Client communication and testing	Completed
Yuqi Wu	Image registration, lesion matching and accuracy evaluation	Pipeline integration and progress reporting	Completed

The table above summarises each member's primary technical and team contributions. Supporting evidence is available through the corresponding Jira tasks, GitHub commits/files, meeting minutes, and project documentation.

Work	Assignee	Reporter	Priority	Status	Resolution	Created	Updated	
P31-21 Display Lesion Matching Details	YUQI WU	YUQI WU	Medium	Done	Done	Sep 07, 2026, 5:41 PM	Sep 12, 2026	
P31-19 Integrate Lesion Matcher into Dashboard	linkang	YUQI WU	Medium	Done	Done	Sep 07, 2026, 5:38 PM	Sep 11, 2026	
P31-18 Evaluate Lesion Tracking Accuracy	YUQI WU	YUQI WU	Medium	Done	Done	Sep 07, 2026, 5:37 PM	Sep 09, 2026	
P31-17 Support Coronal and Sagittal Views	Haoxiong Pan	YUQI WU	Medium	Done	Done	Sep 04, 2026, 5:37 PM	Sep 11, 2026	
P31-16 Implement baseline lesion correspondence matcher	YUQI WU	YUQI WU	Medium	Done	Done	Sep 02, 2026, 6:36 PM	Sep 09, 2026	
P31-15 Generate lesion pair costs	Haoxiong Pan	YUQI WU	Medium	Done	Done	Sep 02, 2026, 6:36 PM	Sep 09, 2026	
P31-14 Extract lesion features for matching	Qiang li	YUQI WU	Medium	Done	Done	Sep 02, 2026, 6:35 PM	Sep 08, 2026	
P31-13 Validate longitudinal scan alignment	YUQI WU	YUQI WU	Medium	Done	Done	Sep 02, 2026, 6:35 PM	Sep 04, 2026	
P31-12 Align baseline and follow-up scans	YUQI WU	YUQI WU	Medium	Done	Done	Sep 02, 2026, 6:34 PM	Sep 04, 2026	
P31-10 Inspect expert lesion correspondence for Cohort A	YUQI WU	YUQI WU	Medium	Done	Done	Aug 24, 2026, 7:34 PM	Sep 02, 2026	
P31-9 Visualise PET/CT and lesion overlays	Qiang li	YUQI WU	Medium	Done	Done	Aug 24, 2026, 7:34 PM	Sep 01, 2026	
P31-8 Load lesion masks and extract individual lesions	Haoxiong Pan	YUQI WU	Medium	Done	Done	Aug 24, 2026, 7:33 PM	Aug 31, 2026	
P31-7 Load baseline and follow-up PET/CT volumes	Haoxiong Pan	YUQI WU	Medium	Done	Done	Aug 24, 2026, 7:32 PM	Aug 31, 2026	

Figure 3. Specific work assignment in Jira

6 Feasible Plan and Risk Mitigation Strategy for the Next Project Stage

The next project stage will focus on improving the reliability of the existing end-to-end pipeline rather than introducing unrelated functionality. Since the main workflow for data preparation, lesion extraction, registration, matching, evaluation, and visualisation is already implemented, the remaining work is considered feasible within the project timeline.

The highest-priority task is to investigate the rigid-registration initialisation issue currently affecting four Cohort A patients. Once corrected, these cases will be reprocessed and the full evaluation will be repeated to avoid selection bias. The team will also investigate known lesion-extraction and ground-truth mapping issues, as these upstream errors can directly affect matching accuracy.

After the pipeline is stable, the matching parameters will be further validated, including distance, lesion-size, unmatched-lesion, and merge-related costs. Sensitivity testing will be used to determine whether the current performance is robust to reasonable parameter changes rather than dependent on a single configuration. The tracking outputs will also be integrated more closely with the Streamlit interface so that correspondence results can be reviewed alongside registered images and lesion masks.

Period	Planned Work	Main Risk	Mitigation
Weeks 7–8	Fix registration and lesion-extraction issues; repeat failed Cohort A cases	Registration or extraction failures remain unresolved	Isolate failing cases, preserve logs/intermediate outputs, and validate visually before downstream processing
Weeks 9–10	Re-run full evaluation and perform matching parameter sensitivity tests	Current accuracy may not generalise beyond the existing subset	Evaluate all available valid cases and report both successful and failed cases transparently
Weeks 11–12	Complete Streamlit integration, reproducibility	Integration or data	Keep modules

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	testing, and where feasible extend testing to Cohort B/PET data	differences may introduce new failures	independent, reuse existing intermediate formats, and prioritise core Cohort A functionality if time becomes limited
Week 13	Final documentation, client demonstration, report, and delivery package	Late integration or documentation delays	Freeze major functionality before final delivery and reserve the final stage for testing, documentation, and presentation

A further project risk is scope expansion. Additional visualisation or analysis features may be useful, but they should not compromise completion of the agreed core requirements. Therefore, registration reliability, lesion correspondence, quantitative evaluation, reproducibility, and the client-facing review workflow will remain the priority. Optional extensions will only be implemented after the core pipeline is stable.

The final delivery will include a reproducible Python pipeline, the Streamlit review interface, evaluation results, source code, dependency and execution instructions, and documentation of known limitations. This staged plan ensures that the remaining work directly addresses the most important technical risks while maintaining a feasible path toward completing the project within the agreed scope and schedule.

7 Reflections and Conclusions

The project has progressed from separate image-processing experiments to an integrated and evaluable lesion-tracking baseline. The modular architecture, patient manifest, intermediate data formats, automated tests, and regular client feedback helped the team integrate work produced by different members.

An important technical lesson was that longitudinal lesion matching cannot rely on voxel indices alone. Image geometry and physical coordinates must be preserved, and registration must occur before distance-based matching. The team also learned that expert correspondence data must be separated from algorithm input to ensure that the evaluation remains meaningful.

The current quantitative results are encouraging, but they should not be interpreted as final full-cohort performance. Four patients remain excluded because of a known registration problem, PET availability is limited in parts of the current sample, and rigid registration cannot represent every local anatomical change.

Overall, the project has delivered a functional end-to-end baseline for lesion extraction, alignment, matching, evaluation, and visualisation. The next stage will prioritise registration reliability, full-cohort evaluation, parameter validation, and stable integration into the final client-facing interface.

Appendix

- **Repository:** [COMP3888 M17 ZL 03 P31 GitHub repository](#)
Primary branch: main
Implementation language: Python
Main dependencies: pandas, NumPy, NiBabel, SciPy, Matplotlib, ITKElastix, Streamlit, NetworkX and pytest
- The repository contains separate tools for preparing the patient subset, extracting lesions, registering longitudinal scans, generating aligned lesion features, calculating pair costs, running min-cost-flow matching, evaluating correspondence accuracy, and launching the visualisation interface. Detailed command-line instructions and expected input/output formats are provided in the repository README and the documents under the [docs](#) directory.
- Patient imaging data are not stored in the public source-code repository. Users must configure the local data root and patient manifest before running the pipeline.
- Jira: [COMP3888_M17_03_ZL](#)

References:

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ITKElastix Developers. *ITKElastix Documentation*.
NetworkX Developers. *NetworkX Network Flow Algorithms*.
Streamlit. *Streamlit Documentation*.
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